

8th CII 3M Compétition 2020

A Kaizen Presentation By

(Everest Industries Ltd)

Podanur Works - Coimbatore

18th September 2020

COMPANY OVERVIEW

Everest Industries Limited, incorporated in 1934, has a rich history in manufacturing of Building products and Steel products. Everest offers a complete range of roofing ceiling, wall, flooring and cladding products distributed through a large network, and also pre-engineered steel buildings for industrial, commercial and residential applications. It is one of the leading building solution providers in India, providing detailed technical assistance in the form of design, drawings and implementation for every project.

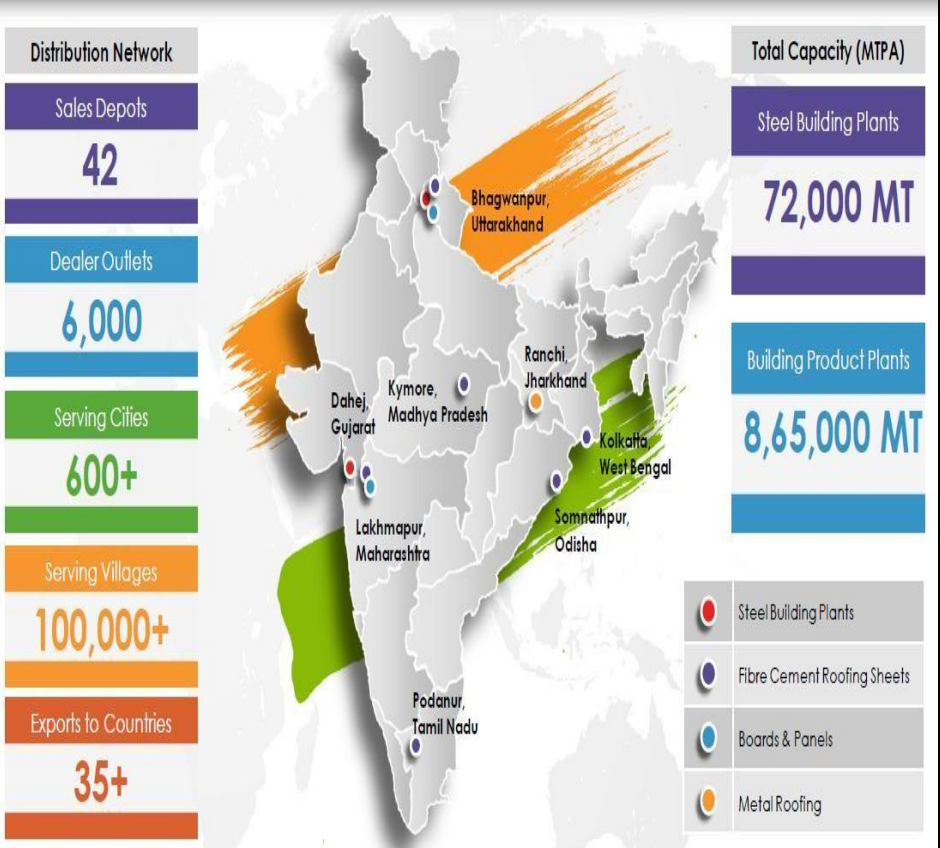
OUR BUSINESS

The company businesses are divided into two distinct business segments with capacity –

- 1. Buildings Products – 8,65,000 MT / Annum
- 2. Steel Buildings – 72,000 MT / Annum

Buildings Products segment contributed 63% of the company’s revenue and includes Fiber cement roofing sheets, Fiber cement boards, Rapicon wall & their accessories.

Steel Building segment accounted for 37% of annual revenue. In this segment Company offers customized building solutions in steel such as Pre-Engineered steel Buildings, Smart steel buildings, Metal roofing and Cladding.



Products:



Fibre Cement Roofing Sheets



Hi-Tech Roofing Sheets



Super Sheets



Cement Planks



Fibre Cement Boards



Rapicon Walls



Smart Steel Buildings



Customers:



Kaizen Theme: To reduce workmen fatigue in De-piling Machine.

Implemented Area: Line No -1, Podanur Works, Coimbatore

Implementation Date: 10th December 2019

Cross Functional Team:



Mr. Y. Kesavaraj
(Prod. Line Incharge)



Team Leader
Mr. C. Murugesan
(Maintenance HOD)



Mr. K. Karthikeyan
(Maintenance)



Mr. N. Anil Kumar
(Engineer - QS & BE)



Mr. Pillai
(Fabricator)



Mr. A. Devaraja
(Fitter)

Theme

To eliminate workmen fatigue in De-piling machine.

Reason for selecting THEME :

- *Continuous workload on De-piling workmen for aligning every sheet in stack.*
- *Sheet aligning is more time-consuming & repetitive work.*
- *During small breaks, 1 person aligns sheets on both sides.*
- *Very less time for doing other De-piling work activities.*

Problem :

- Workmen fatigue in De-piling machine.

Workmen has to align each and every sheet in stack continuously without any delay. And simultaneously, workmen has to check the visual quality of each sheet, remove defective sheets from the stack & to do small finishing work on sheets in a span of 14 seconds. In this, sheets aligning is consuming more time & effort.

Process Flow:



Problem :

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1. Sheets Trolley



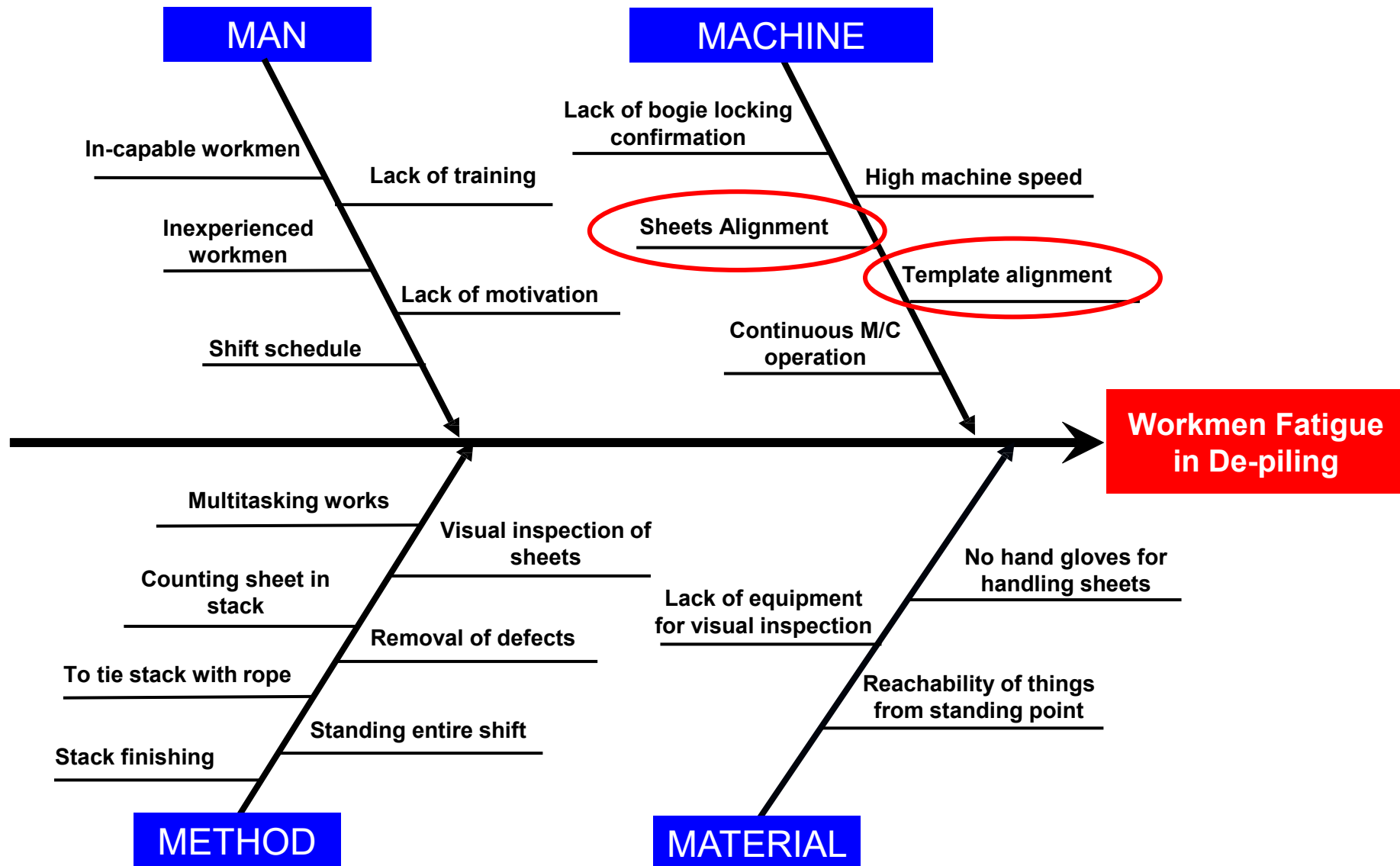
2. Sheet Separation pad

- 1) Workmen has to place template on the trolley & adjust till it is perfectly aligned with Sheet Separation(SS) pad.
- 2) Every sheet is adjusted manually on stack by workmen. It leads to workmen fatigue.



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Cause & Effect Diagram (Analysis)



Problem: Workmen fatigue for aligning sheets

& templates.

Why 1

Due to manual adjustment of sheets in stack.

Manual adjustment of templates on trolley.

Why 2

Sheets are not separated & aligned by Sheet separation pad.

Template not aligned with SS Pad.

Why 3

Inadequate design of SS Pad to align & pile sheets in the stack automatically.

No fixtures on trolley for positioning templates.

Why 4

Sheet separation pad is a single pad design.

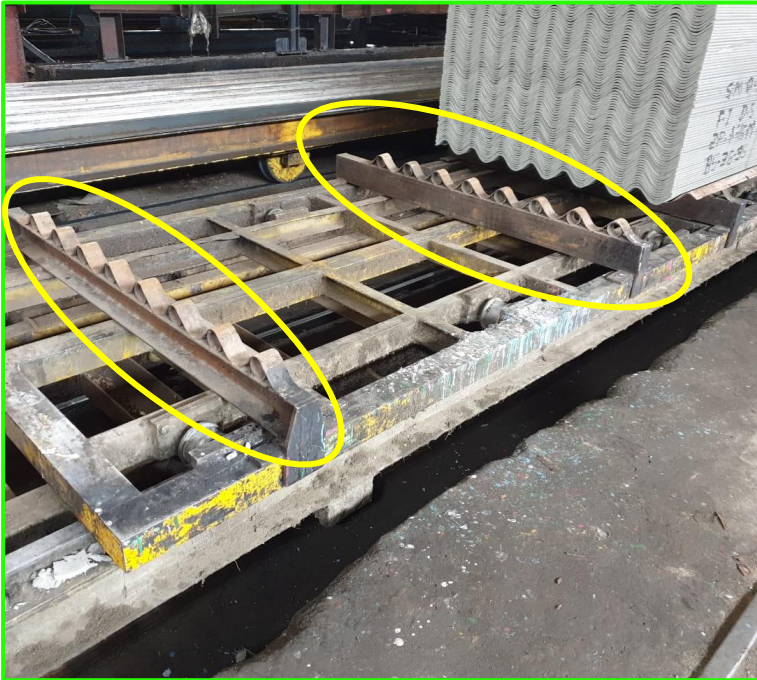
Root Cause:

- 1) Sheet separation pad is a single pad design, which can't align & pile the sheets in stack.
- 2) No fixtures on trolley for positioning of templates.



Idea Generation: To build auto sheet separator with 2 stacking pads as in-house project. And to provide fixtures for easy positioning of template on trolley.

Steps involved in kaizen development					
S.No	Steps	Start Date	Target Date	Responsibility	Status
01	To provide fixtures on sheet trolley for easy positioning of templates.	1/09/2019	14/09/2019	Maintenance Department	Completed
02	To make measurements of new SS pad dimensions as per De-piling requirement.	01/09/2019	03/11/2019	Maintenance Department	Completed
03	To built new SS pad as per the machine requirements.	08/11/2019	08/11/2019	Maintenance Fabricator	Completed
04	To install new SS pad to the De-piling machine.	09/12/2019	10/12/2019	Maintenance Department	Completed
05	To install sensors for PLC control.	09/12/2019	10/12/2019	Electrical Department	Completed



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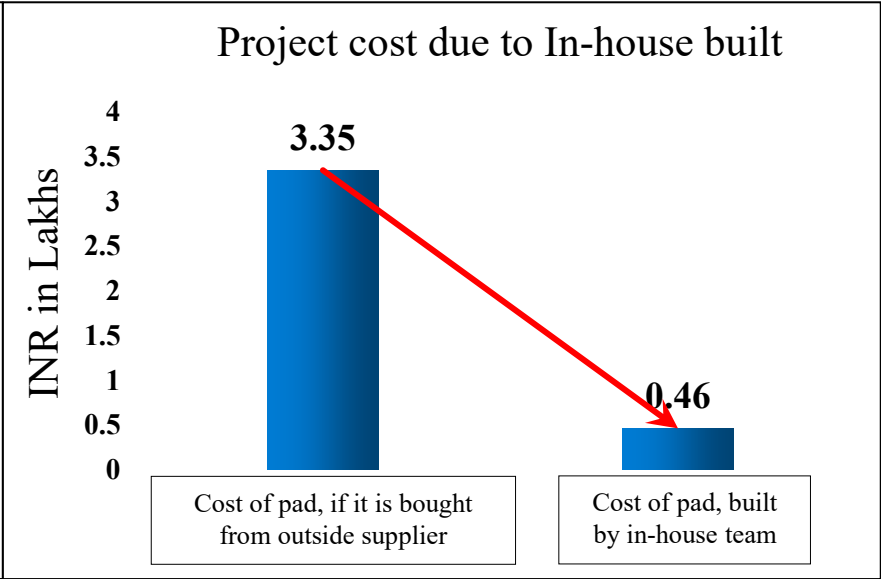
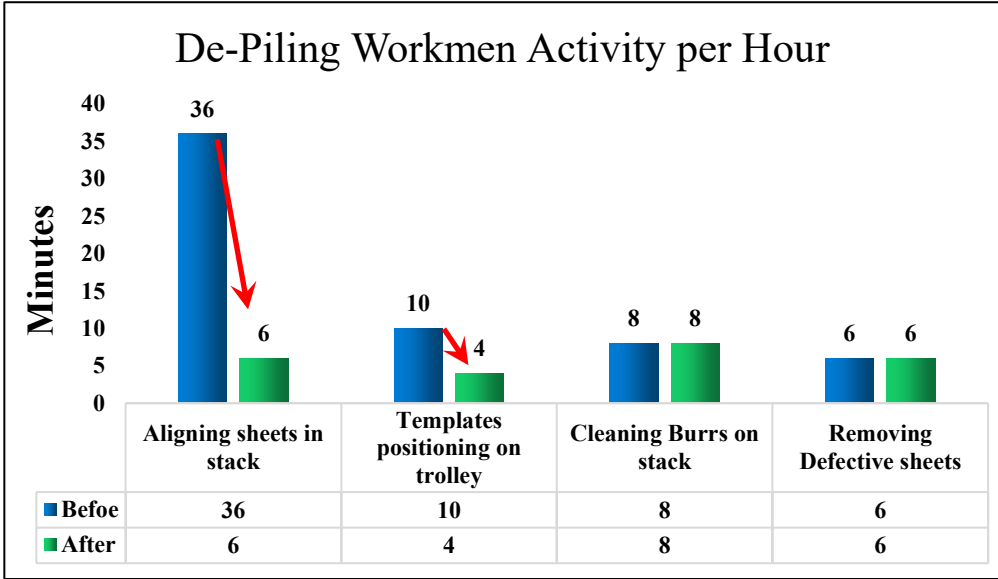


2. Sheet Separation pad

- 1) Fixture which is aligned with SS Pad provided for placing template on it. So, template adjustment is not required.
- 2) Sheet Separation pad separates each sheet & aligns them in stack without workmen intervention.



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Post Lockdown Benefit : While we started Machine after Covid-19 restrictions, this system became very useful. As daily new manpower used to come for work. So, it even became very advantageous for them to work with less effort & with less experience.

Benefits :

- 1) **Productivity** - No Delay in De-piling machine. Due to this, there is no delay in Piling machine, as both M/C's processes are interlinked.
- 2) **Quality** - Sheet Quality inspection has improved in De-piling, as there is no need to manually adjust sheets.
- 3) **Cost** - Cost saving of INR. 2.89 Lakhs for project due to in-house built.
- 4) **Safety** - No Safety issues as there is less human interference.
- 5) **Morale** - Morale of workmen high due to reduction of **Muri**.

Validation of Kaizen in 7 Wastes of Manufacturing

T	Transportation	NA
I	Inventory	NA
M	Motion	Effort for Manual alignment of sheets reduced (↓) by 83%.
W	Waiting	Total worktime decreased (↓) from 60 minutes to 24 minutes/hour
O	Over-Production	NA
O	Over-processing	Breakages due to manual handling of sheets has reduced.
D	Defects	Improvement in Visual Quality inspection due to decreased workload.

- Training on Updated Work Instructions are given to all the workmen for regular inspection & maintenance.

	EVEREST INDUSTRIES LIMITED PODANUR WORKS IS/ISO 9001 : 2015 WORK INSTRUCTION	DOC NO : EL/WI/528 REV NO : 01 PAGE : 1 OF 1 DATE : 11/12/2019
	Work Instruction: WI 528	
	DE-PIILING WORKMEN :	
	Procedure: 1. Place rope across the width of the sheets trolley (i.e. below template fixtures) for tying the stack, once stack is full. 2. Place 2 templates on each fixture of the sheets trolley. 3. Let the sheet separation pad separated the sheets, aligns & stacks the sheets on the sheets trolley. 4. Continuously monitor the quality fo sheets & if any defective sheets are found while de-piling sheets, reject & takeout the sheets from the stack. 5. Place the rejected sheets in a rejection sheets area. 6. If there is any minor finishing defects on sheets, repair the sheets immediately. 7. Check frequently the count of sheets in stack & once the stack is filled i.e.95 sheets/stack, change it to new trolley. 8. Write Details of the stack i.e. Qty of sheets, batch code, shift incharge, de-piling operator, fork lift operator name & quality condition of stack. 9. Now, tie the stack with rope for shifting stack to curing yard by fork lift.	

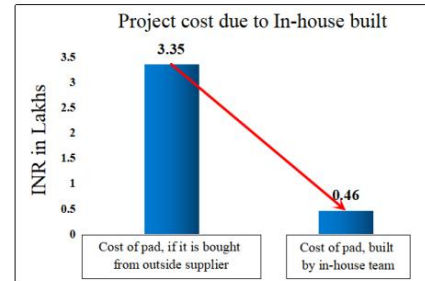
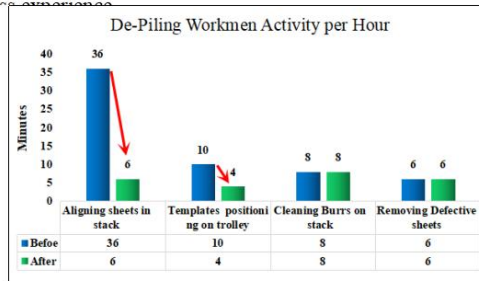
Kaizen Sustenance:

WHAT TO DO	Regular Inspection & Maintenance.
HOW TO DO	Regular Training on Standardized work instructions
FREQUENCY	Weekly

Horizontal Deployment:

Line No.1	Line No. 2
Implemented	Implemented

8th Edition of 3M Competition 	3M KAIZEN IDEA		Activity (Please select any one category only)	Muda	Mura	Muri	 Everest Industries Limited	
	Muda/Mura/Muri					✓		
	Tick wherever necessary		Benefit Area	P ✓	Q ✓	C ✓		D
Plant : Podanur, Coimbatore		Area/Line/Machine : Shop Floor						
Kaizen Theme : To reduce workmen fatigue in De-Piling Machine.		Idea (Logical correlation with root cause) : To built Auto Sheet separator & stacking pad as in-house project. And to provide fixtures for easy positioning of template on trolley.						
Problem/Present Status : Workmen fatigue in De-Piling machine.	Countermeasure (Engineering solution):				Base Data		Manual adjustment of sheets & templates. High Fatigue.	
	Before :		After :		Target		Auto adjustment of sheets & templates. High Morale.	
					Kaizen Start		01-09-2019	
					Kaizen Finish		10-12-2019	
					Team Members		1) C. Murugesan - Maintenance 2) K. Karthikeyan - Maintenance 3) A. Devaraja - Fitter 4) Pillai - Fabricator	
					Benefits: (P,Q,C,D,S,M)		1) P - No Delay in De-piling machine. Due to this, there is no delay in Piling machine, as both M/C's processes are interlinked. 2) Q - Sheet Quality inspection has improved in De-piling, as there is no need to manually adjust sheets. 3) C - Cost saving of INR. 2.89 Lakhs for project due to in-house built. 4) S - No Safety issues as there is less human interference 5) M - Morale of workmen high due to reduction of Muri.	
					Sustenance: Updated SOP of De-Piling machine.			
					Scope & Plan for Horizontal Deployment			
				SN	M/c	Target date	Responsibility	Status
				1	Line 2	01-02-2020	Maintenance HOD	Completed



Thank You